INCH-POUND

MIL-DTL-32746(AV)
4 June 2024

DETAIL SPECIFICATION

# FILTER ELEMENTS, FLUID PRESSURE, HYDRAULIC LINE, 14 MICRON (C) AND 6 MICRON (C), TYPE II SYSTEMS, GENERAL SPECIFICATION FOR

This specification is approved for use by
all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers hydraulic filter elements which retain all particles larger than 14 microns (c) and 6 microns (c) and are suitable for use as specified in 6.1 and the applicable specification sheets.

Comments, suggestions, or questions on this document should be addressed to the Commander, U.S. Army Futures Command, DEVCOM Aviation and Missile Center, Attn: Technical Data Management Division (TDMD), 5687 Logistics Road, Building 5687, Redstone Arsenal, AL 35898-5000, Systems Readiness Directorate (SRD), Redstone Arsenal, AL 35898-5000, or email: joseph.m.casamatta.civ@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 1650

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1.2 Classification. Hydraulic filter elements are either cleanable or non-cleanable as specified herein and in the applicable specification sheet.

1.2.1 Temperature.

| | | |
|---|---|---|
| Type I | – | -65 ºF to +160 ºF Fluid temperature range as defined in SAE-AS5440 |
| Type II | – | -65 ºF to +275 ºF Fluid temperature range as defined in SAE-AS5440 |

1.2.2 Pressure.

| | | |
|---|---|---|
| Class 1500 | - | Filter and filter element furnished in accordance with this specification is suitable for operating with pressures up to and including 1,500 PSI as defined in SAE-AS5440. |
| Class 3000 | - | Filter and filter element furnished in accordance with this specification is suitable for operating with pressures up to and including 3,000 PSI as defined in SAE-AS5440. |
| Class 4000 | - | Filter and filter element furnished in accordance with this specification is suitable for operating with pressures up to and including 4,000 PSI as defined in SAE-AS5440. |
| Class 5000 | - | Filter and filter element furnished in accordance with this specification is suitable for operating with pressures up to and including 5,000 PSI as defined in SAE-AS5440. |

1.2.3 Styles.

Style A – Bypass housing incorporates provision for bypassing fluid at 100 +/- 10 psi.

Style B – Non-Bypass housing no provisions for bypassing fluid.

1.2.4 Tube sizes and micron ratings. Tube sizes and micron ratings are as specified herein and in the applicable specification sheet.

1.3 Part or Identifying Number (PIN). The part or identifying numbers (PIN) are specified in the applicable specification sheets.

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has



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been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, and 5 of this specification, whether they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

| | | |
|---|---|---|
| MIL-DTL-117 | - | Bags, Heat-Sealable |
| MIL-STD-130 | - | Identification Marking of U.S. Military Property |
| MIL-PRF-680 | - | Degreasing Solvent |
| MIL-STD-810 | - | Environmental Engineering Considerations and Laboratory Tests |
| AN929 | - | Cap Insert Assembly, Pressure Seal, Flared Tube Fitting |
| MIL-PRF-5606 | - | Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance (Inactive for new design) |
| MIL-DTL-8815 | - | Filter, Fluid, Pressure, Hydraulic Line, 3,000 PSI, Absolute 15 micron Absolute and 5 Micron Absolute, Type II Systems, General Specification For |
| MIL-PRF-81836 | - | Filter and Disposal Element, Fluid Pressure, Hydraulic 3 Micron Absolute, General Specification for |
| MIL-PRF-83282 | - | Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Metric, NATO Code Number H-537 |
| MIL-PRF-87257 | - | Hydraulic Fluid, Fire Resistant, Low Temperature, Synthetic Hydrocarbon Base, Aircraft and Missile |
| MS21344 | - | Fittings, Installation of Flared Tube, Straight Threaded Connectors, Design Standard For |

(See supplement 1 for list of complete specification sheets)

(Copies of these documents are available online at https://quicksearch/dla.mil.)



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2.2.2 Other government documents, drawings, and publication. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

INTERNATIONAL STANDARIZATION ORGANIZATION (ISO)

| | | |
|---|---|---|
| ISO 2941 | - | Filter elements - Verification of Collapse/Burst Pressure Rating |
| ISO 3968 | - | Hydraulic fluid power – Filters – Evaluation of differential pressure versus flow |
| ISO 4406 | - | Hydraulic Fluid Power – Fluids – Method For Coding The Level Of Contamination By Solid Particles |
| ISO 11171 | - | Hydraulic Fluid Power – Calibration Of Automatic Particle Counter For Liquid (DoD adopted) |
| ISO 12103 | - | Road vehicles - Test Contaminants for Filter Evaluation |
| ISO 16889 | - | Hydraulic fluid power — Filters — Multi-pass method for evaluating filtration performance of a filter element |

(Copies of this document are available for purchase online at: https://www.iso.org.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

| | | |
|---|---|---|
| SAE-AS598 | - | Aerospace Microscopic Sizing and Counting of Particulate Contamination for Fluid Power Systems (DoD adopted) |
| SAE-ARP785 | - | Procedure for the Determination of Particulate Contamination in Hydraulic Fluids by the Control Filter Gravimetric Procedure (DoD adopted) |
| SAE-AIR787 | - | Filter Element Cleaning Methods |
| SAE-ARP901 | - | Bubble Point Test Method (DoD adopted) |
| SAE-ARP1383 | - | Impulse Testing of Hydraulic Components (DoD adopted) |



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| | | |
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| SAE-AS4059 | - | Aerospace - Cleanliness Classification for Hydraulic Fluids |
| SAE-AS5174 | - | Fitting, Union and Reducer, Flared (DoD adopted) |
| SAE-AS5440 | - | Hydraulic Systems, Military Aircraft, Design and Installation Requirements |
| SAE-AS8775 | - | Hydraulic Systems Components, Aircraft and Missiles, General Specification for (DoD adopted) |

(Copies of these documents are available for purchase online at www.sae.org.)

AEROSPACE INDUSTRY ASSOCIATION (AIA)

| | | |
|---|---|---|
| NAS847 | - | Caps and Plugs, Protective, Dust and Moisture Seal (DoD adopted) |

(Copies of these documents are available for purchase online at www.aia-aerospace.com.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

| | | |
|---|---|---|
| ASME-Y14.5M | - | Dimensioning and Tolerancing (DoD adopted) |
| ASME-Y14.38 | - | Abbreviations and Acronyms For Use In Product Definition And Related Documents |

(Copies of this document are available for purchase online at www.asme.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

| | | |
|---|---|---|
| ANSI/ASQ Z1.4 | - | Sampling Procedures and Tables For Inspection By Attributes (DoD adopted) |

(Copies of this document are available for purchase online at www.asq.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 Specification sheets and classification of inspections. The individual requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of conflict between the requirements of this specification and the specification sheet, the latter shall govern.



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3.2 Qualification. The filter elements furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable Qualified Products List (QPL) before contract award (see 4.2 and 6.3).

3.3 General specification and responsibility for inspection. The requirements of SAE-AS8775 apply as requirements of this specification with the exception and additions specified herein. When the two specifications conflict, this specification shall govern.

3.4 Materials.

3.4.1 Compatibility, filter material. The contractor shall select materials capable of meeting all the operational and environmental requirements specified herein. The filter elements shall be constructed of materials that shall not adversely affect or be affected by hydraulic fluid conforming to MIL-PRF-5606, MIL-PRF-83282, MIL-PRF-87257 and other test fluids included herein.

3.4.2 Process controls limitations. If the filter element is of the type wherein resinous or other material is used for elimination of imperfections, not more than 5 percent of the filtering area shall be covered by the applied material. Processes must be written and provided to the procuring activity.

3.4.3 Filter element repair procedures. Filter element repair procedures as allowed in the specification sheet, shall be submitted for approval to the qualifying activity.

3.5 Filter design and construction.

3.5.1 Filter elements, examination of filter elements design and construction. Filter elements conforming to SAE-AS5440 shall be designed and constructed as specified herein and in the applicable specification sheet.

3.5.2 Flow type. Filter elements shall be of the full-flow type. The flow through the filter elements shall be from “outside-in” unless otherwise specified.

3.5.3 Weight. The total weight of the filter shall be as specified in the applicable specification sheet.

3.5.4 Interchangeability. Filter elements shall be designed to prevent installation of filter elements into incorrect filter housings.

3.5.5 Protective devices.

3.5.5.1 Filter element with relief valves. Filter element pressure relief (and reseat) valve, when required by the applicable specification sheet, shall meet the requirements of the applicable specification sheet and the requirements and tests specified herein. In addition the filter element test housing shall have a suitable means of stopping flow through the filter element.



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3.6 Performance.

3.6.1 Filter element. Filter elements shall meet the requirement of this specification, if they perform satisfactorily when subjected to the tests specified herein.

3.6.1.1 Filter element tests. The total quantity of filter elements (noncleanable) required for qualification shall consist of thirteen (13) elements of each size upon which qualification is desired.

3.6.1.2 Filter element media and O-ring. The filter elements and O-ring shall be designed to withstand normal handling and installation.

3.6.1.3 Repaired filter elements. Filters elements that are repaired and procedures approved by the qualifying activity, as allowed in the specification sheet, shall perform satisfactorily to all tests specified herein. These filters will be tested exactly as identified in 4.6.

3.6.2 Test conditions. Filter elements shall perform satisfactorily when subjected to the following tests specified herein.

3.6.3 Bubble point.

3.6.3.1 Bubble point tests. All filter elements shall be tested in accordance with SAE ARP901 and values recorded.

3.6.3.2 Bubble point (initial). The initial bubble point (B/P) value is the B/P established at the beginning of qualification testing recommended by the manufacturer.

3.6.3.3 Bubble point after performance test. As specified in the qualification test procedure of this specification, a B/P test shall be conducted before and after each qualification test. The filter element B/P value shall not be less than the initial B/P value determined in 3.6.3.2 above.

3.6.3.4 Bubble point test, sequence determination, and ranking. Following cold start, each of the 12-filter element shall be B/P tested in accordance with SAE ARP901 and ranked.

3.6.3.5 Minimum and documented bubble point. This B/P value is defined by successfully conducting the filter conditioning tests and determining the lowest B/P of the test samples to occur after Immersion and Thermal Cycling, Pressure Drop, and Cold start test. This value will differ for various filtration media by each filter manufacturer. This minimum B/P value is the lowest B/P value measured after filter conditioning and successfully completing the dynamic efficiency test and shall be documented on the drawing. (See Figure 2)

3.6.4 Test housing and element pressure drop. A filter test housing shall be manufactured with the requirements to meet pressure and flow see Figure 1, as specified in the applicable specification sheet. The filter test housing manufactured and used for qualification shall be used again in subsequent filter qualification testing.



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3.6.5 Filtering media construction, immersion, and thermal cycling. Filter elements shall perform satisfactorily without degradation when subjected to the temperature cycles specified herein.

3.6.6 Cold start. Filter elements shall undergo cold start testing as specified herein.

3.6.7 Filter dynamic efficiency. Filter element efficiency shall be as specified in the applicable specification sheet and shall meet the requirements specified therein.

3.6.8 Filter elements unloading. Filter elements, when fully loaded to a differential pressure of 300 psid +/- 5%, or as specified in the applicable specification sheet, shall not experience a reduction in differential pressure. This differential pressure reduction shall not exceed 10% from the terminal differential pressure when subjected to intermittent hydraulic pulsing or start up flow conditions as found in typical hydraulic systems.

3.6.9 Media migration. Filter elements shall withstand vibration frequencies in accordance with paragraph 4.6.9. There shall be no media migration traceable to filter element, its adhesives, or its packaging.

3.6.10 Flow fatigue. Filter element shall withstand flow fatigue (pressure and flow cycling loading) and exhibit no failure to the filter element inside the test housing.

3.6.11 Pressure element build-up and collapse. Filter elements shall withstand a differential pressure as specified in the applicable specification sheet.

3.7 Markings.

3.7.1 Identification of product. Each filter element shall be clearly and permanently identified on one end by steel stamping, electro-etching, or by applicable laser marking as outlined in paragraph 4.7.

3.8 Conformance inspection. Each filter element shall be inspected consistent with the following:

3.8.1 Individual inspection. The individual inspection shall be conducted on all filter elements in the order specified in Figure 3. Any filter element containing a defect shall be rejected.

3.8.2 Sampling lot inspections. The sampling lot inspection shall be conducted on filter elements in the order specified in Figure 3. Quantity and recurring schedules should be based on vendor quality procedures and processes. Sample lots, containing defective filter elements during inspection shall be reported to the procuring and qualifying activity immediately.



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## 4. VERIFICATION

4.1 Specification sheets and classification of inspections. The individual testing requirements of the specification sheet and classification of inspections shall be as specified herein and as set forth in the specification sheet for that size filter element and be tested as follows:

a. Qualification inspections (Figure 1 and Figure 2).
b. Conformance inspection (Figure 3).

4.2 Qualification. The filter elements, after successfully completing the qualification requirements under this specification and the specification sheet, shall be products which are authorized for use by the qualifying activity and other DoD services. The qualified filter element will be placed on the applicable Qualified Products List (QPL) before contract award. In addition, the retention of the qualification for the filter element on the applicable QPL shall be dependent on periodic verification of continued compliance with the requirements set forth of this specification.

4.2.1 Tests. The qualification tests shall consist of all the tests specified in paragraph 4.6 conducted on the applicable specimens in the order specified in Figure 1 and Figure 2. Any test event that does not meet applicable test requirements shall be considered a "Test Failure" and must be reported to the procuring agency prior to proceeding with additional tests. The qualifying agency reserves the right to witness all tests conducted under this specification with at least 30 days' notice (in writing) prior to test.

4.2.1.1 Qualification test report (QTR). The qualification test report should include the following:

a. The validation test and actual test results as described in paragraph 4.
b. The test reports, submitted as specified and in full knowledge of the US Army, as defined in 4.2.2.
c. The filter element drawing and the applicable production acceptance test procedure (ATP) should include the minimum and documented bubble point (4.6.4.5).
d. All resulting data which includes Figure 10, 11, 12, and 13 which includes: graphs, pressures, flows, vibration levels, digital photographs of media migration, temperature, and particulate capture as defined in paragraph 4. will be presented and documented.
e. Provide the pictures of the tests set-up and schematic shown in Figures 4, 5, 6, 7, 14, and 16.

4.2.2 Source approval request (SAR). Suppliers are required to submit a SAR to U.S. Army Futures Command, DEVCOM Aviation and Missile Center, Attn: Technical Data Management Division (TDMD), 5687 Logistics Road, Building 5687, Redstone Arsenal, AL 35898-5000 prior to beginning the process to qualify as a Qualified Source. The following documentation must be included in the SAR.



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4.2.2.1 Contents of a SAR. The SAR provided by the supplier, shall include: All manufacturing, assembly, and inspection points shall be controlled by a detailed written procedure outlining each step or parameter of the filter manufacturing process along with any materials, tooling equipment, environmental control, and operator certification required that leads to the specific filter production. Plans shall clearly identify all critical materials, processes and procedures and will include identification in accordance with contractor procedures, as to its revision. All process plans shall clearly define sequences of operation, machine type, and accept/reject limits for the specific process or operation. Any critical processes not easily verified by subsequent inspection shall be clearly defined in the process operation with tolerances.

4.2.3 Retention of qualification. The retention of qualification shall consist of periodic verification to determine compliance of the qualified filter element with the full requirements of this specification and applicable specification sheet. Verification shall be conducted at intervals not exceeding 4 years. In addition, full qualification documentation to show compliance to the tests specified in Figure 2, shall be provided every 4 years to the procuring agency to retain qualification on the Qualified Products Database (QPD). Without this full requalification, the procuring agency will remove the company from the QPD as appropriate.

4.2.4 Waivers and requests for relief. During requalification, a waiver for relief in conducting specific tests can be requested in writing, from the qualification authority. The vendor must show previous testing had been satisfactorily completed during periodic lot sampling within the contract period of performance. Approved waivers will detail exceptions in writing; however, the dynamic efficiency test must be performed during requalification and cannot be waived. Vendors shall not be granted two consecutive waivers.

4.2.5 Requalification due to changes. Any qualified filter under this specification will immediately require requalification due to the following activities: change in media or materials, change in manufacturing processes, re-opening of the contract, and company buyout (change of ownership).

4.3 General specification and responsibility for inspection. The quality assurance provision shall be in accordance with SAE-AS8775 and as specified herein. Unless otherwise specified in the contract, the contractor is responsible for the performance of all testing and inspection. Except as otherwise specified in the contract, the contractor may use their own or any other Army approved facilities suitable for the performance of the inspection and testing requirements specified herein, and as approved by the procuring agency. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Additionally, the Government reserves the right to witness any testing / inspections set forth in this specification with at least 30 days' notice (in writing) prior to test.

4.4 Materials.

4.4.1 Compatibility, filter material. The filter elements material shall be conforming to SAE-AS5440 and as specified herein and in the applicable specification sheet. The filter element material shall be capable of meeting all the operational and environmental requirements



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specified herein and on the specification sheet. The filters shall be constructed and tested and shall not be adversely affected by hydraulic fluid conforming to MIL-PRF-5606, MIL-PRF-83282, MIL-PRF-87257 and other test fluids included herein.

4.4.2 Process controls limitations.

4.4.2.1 Repair process controls limitations. If the filter element is of the type wherein resinous or other material is used for elimination of imperfections, not more than 5 percent of the filtering area shall be covered by the applied material. Filter element media having an initial B/P of less than 5-inches of water before repair shall be cause for rejection. The manufacturer's qualification test report shall define the resin employed for joining or patching, application techniques, curing cycle, age limitation, and procedures followed to insure adhesion of the resin. The manufacturer's drawing shall specify the limitations or process controls that shall govern manufacturing variations in media obstruction due to braze wicking, seam width, and crimp length in addition to the existing limits on repair.

4.4.3 Filter element repair procedures. Filter elements that can be repaired shall have the repair procedures submitted for approval to the qualifying activity responsible for qualification with the qualification test report. These repair procedures shall specify but shall not be limited to the resin employed or welding for joining or patching, application techniques, curing cycle, procedures to be followed to ensure adhesion of the resin and special tools and test equipment required. All qualification tests contained within this specification must be completed and satisfactorily pass qualification after the repair is made to ensure compliance of this document and that of the specification sheet.

4.5 Filter design and construction.

4.5.1 Filter elements, examination of filter elements design and construction. The filter element shall be designed and constructed to SAE-AS5440 and carefully examined to determine conformance to the requirement of this specification for design, weight, workmanship, marking, conformance to applicable AN or MS standard, government and manufacturer's drawings, specifications sheets and standards for any visible defects or damage that may have occurred during manufacturing, handling, movement or testing of the elements.

4.5.2 Flow type. Filter elements shall be of the full-flow type. The flow through the filter elements shall be from "outside-in" unless otherwise specified on the applicable specification sheet.

4.5.3 Weight. The total weight of the filter shall be as specified in the applicable specification sheet.

4.5.4 Interchangeability. Filter elements shall be designed to prevent installation of filter elements into incorrect filter housings. The correct part number of the filter element shall be verified prior to installation into the test filter housing.



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4.5.5 Protective devices.

4.5.5.1 Filter elements with relief valves. Filters requiring pressure relief valve, when required by the applicable specification sheet, shall be tested, and meet the requirements of the applicable specification sheet and the requirements and tests specified herein. The filters must pass the entire spectrum of tests identified in Figure 2, in the order presented. In addition, a suitable means of stopping flow through the filter element port or the test housing shall be installed for this type of element only.

4.5.5.2 Relief valve operation (if applicable). Filter elements having integral relief valves, see applicable specification sheet, shall be subjected to the following tests. The filter element media shall be blocked, and the filter element installed in the test housing. By means of a power-driven pump, pressure shall be applied to the inlet port of the filter housing, beginning with a pressure of 20 psi and increasing in increments of 10 psi or less until cracking pressure of the relief valve in the filter element being tested is reached. At each increment, the pressure shall be maintained constant for 5 minutes. At each pressure increment, time shall not be considered "in" or "leakage" noted until the beginning of the third minute. The leakage rate during the last 3 minutes of each pressure increment shall be noted. The pressure at which the leakage rate through the relief valve amounts to 8 cubic centimeters (cc) approximately 160 drops per minute, shall be considered as cracking pressure. Cracking pressure shall occur at 50 ± 5 psi or as required per the applicable specification sheet. Leakage up to 80 percent of the cracking pressure shall not exceed 4 cc. (approximately 80 drops) per minute. Pressure shall be increased until rated flow through the relief valve is obtained. Rated flow shall occur at a pressure differential between inlet and outlet ports not exceeding 80 psi or as required per the applicable specification sheet. The pressure shall be reduced in 10 psi increments as above until leakage through the relief valve does not exceed the rate of 10 drops per minute. The resulting pressure shall be considered the relief valve reseating pressure and shall occur at a pressure of 30 psi or as required per the applicable specification sheet.

4.6 Performance.

4.6.1 Filter elements. The filter element shall satisfactorily perform all testing specified herein or as required in the applicable specification sheet. The filters must pass the entire spectrum of tests identified in Figure 2, in the order presented.

4.6.1.1 Filter element tests. Filter elements shall consist of thirteen (13) elements of each size upon which qualification is desired. Of these elements, ten (10) shall undergo the successful completion of test inspections outlined in Figure 2. Three (3) of these elements will be designated for use as spares, intended to be used if damage is incurred on other elements outside of the test environment. One (1) element will undergo Media Migration test based upon recent manufacturing processes and packaging and will not be B/P tested, per Figure 2. Manufacturer's qualification tests shall not have been performed on any samples prior to submission for testing.

4.6.1.2 Filter element media and O-ring. Filter elements and O-ring shall be installed and removed by hand, six times into the filter test housing. The filter elements and O-ring shall be inspected such that under normal handling and installation of the filter, the filter media and O-



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ring shall not be damaged or degraded. One photograph shall be taken of the filter element media and O-ring after testing and be documented in the qualification test report.

4.6.1.3 Repaired filters elements. If the elements are of the type in which repair material is applied to eliminate imperfections, the qualification test elements shall have a maximum of 5 percent of the filtering area repaired. The areas repaired shall contain imperfections whose initial B/P is no less than 5 inches of water. Repaired filters must successfully pass testing as outlined in paragraph 4.6 and Figure 2. This repair is allowed after use of the three spare filters to continue testing.

4.6.2 Test conditions. Filter elements shall perform satisfactorily when subjected to the following fluid test conditions.

4.6.2.1 Test fluid. Unless otherwise specified, the hydraulic fluid used for all tests shall conform to MIL-PRF-5606, MIL-PRF-83282, or MIL-PRF-87257 as applicable. There shall be no free water in fluid used for testing. Fluid sampling shall be documented from an inhouse calibrated instrument or an appropriate ISO Certified laboratory and maintained for future reference.

4.6.2.2 Test fluid cleanliness. For conformance tests and critical qualification tests, the test fluid shall be pre-cleaned to a cleanliness level equal to or better than ISO 4406 15/13/10 Cleanliness Level (>4 um / >6 um / >14 um) micron (c) method. Fluid sampling shall be documented from an inhouse calibrated instrument or an appropriate ISO Certified laboratory and maintained for future reference.

4.6.2.3 Test fluid filtration. Unless otherwise specified, the test fluid shall be continuously filtered through a filter with 99.5% efficiency per ISO 16889 or higher. The cleanliness of this fluid shall be substantiated with a particle counter and documented.

4.6.2.4 Test fluid temperature. Unless otherwise specified, the actual temperature of the test fluid shall be 100ºF to 160ºF ±15ºF or as specified in the specification sheet for applicable test.

4.6.2.5 Ambient temperature. Unless otherwise specified, the tests shall be conducted at a room temperature of 70ºF to 90ºF measured within 12 inches of the test sample.

4.6.2.6 Test dust and source of supply. Unless otherwise specified by the qualifying activity, test dust use in the performance of filter or filter element testing shall be ISO medium test dust. Test dust used for qualification tests shall be obtained from Powder Technologies, Inc. 1300 Grey Fox Road, Arden Hills, MN 55112, or other sources approved per ISO 12103-1 A3.

4.6.3 Bubble point.

4.6.3.1 Bubble point tests. Each filter element shall be individually tested to determine the B/P. The filter element shall be tested and documented in accordance with SAE ARP901.



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4.6.3.2 Bubble point (initial). This initial B/P value is established at the beginning of qualification testing recommended by the Manufacturer. This value will differ for various filtration media by each filter manufacturer. No filter element with a B/P value below 5-inches water column will be accepted.

4.6.3.3 Bubble point after performance test. As specified in the Qualification Test Procedure of this specification, a B/P test shall be conducted before and after each qualification test as specified in Figure 2. The filter element B/P value shall not be less than the initial B/P value determined in 4.6.3.2.

4.6.3.4 Bubble point test, sequence determination, and ranking. One filter element shall be tested for media migration in the test filter housing. Following cold start, each of the remaining 12-filter elements as shown in Figure 2, shall be B/P tested in accordance with SAE ARP901. Bubble point values shall be recorded on twelve (12) filter elements. The outcome of this test will also determine which test sequence each filter element will follow, as outlined in Figure 2. The filter element rankings and sequences are as follows: seven of the lowest elements (7 articles) with the lowest B/P rankings, three of which shall perform the dynamic efficiency sequence and three (3), set aside as spares (in case of damage or premature failure). The seventh ($7^{th}$) element with the lowest B/P ranking (1-article) shall undergo the flow fatigue sequence. The two elements with the highest B/P rankings (2-articles) shall undergo the collapse sequence. The remaining elements (3-articles) shall undergo the pressure drop sequence tests.

4.6.3.5 Minimum and documented bubble point. This B/P value is defined by successfully conducting the filter conditioning tests and determining the lowest B/P of the test samples to occur either after immersion and thermal cycling, pressure drop, or after cold start. This value will differ for various filtration media by each filter manufacturer. This minimum B/P value is the lowest B/P value measured after successfully completing the dynamic efficiency test and shall be documented on the drawing. This B/P value shall be used in future acquisitions for acceptance by the qualifying activity.

4.6.4 Test housing and element pressure drop.

4.6.4.1 Element pressure drop (initial). Twelve (12) of the thirteen (13) filter elements as listed in Figure 2, shall undergo an initial differential pressure test measuring the differential pressure upstream and downstream of the filter under test using a differential pressure transducer or two pressure transducer gauges as specified in ISO 3968, at 100°F ± 5°F using MIL-PRF-83282. Test fluid cleanliness during this test shall be IAW paragraph 4.6.2.2. The pressure drop across the element at rated flow shall not be greater than specified in the applicable slash sheet. The pressure drop shall be recorded for each element. The pressure differential between the Pre-Cold Start Test and the Post-Cold Start Test shall not exceed 25%. Note that while the initial pressure drop values are recorded for each filter element, only three (3) elements will undergo a secondary pressure drop sequence following cold start as specified in Figure 2.

4.6.4.2 Element pressure drop (second). Each element in this sequence shall undergo a (second) pressure drop test per 4.6.4.1, after cold start test. The pressure differential between the initial pressure drop values and this pressure drop shall not exceed 25%. Note that while the



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initial pressure drop values are recorded for each filter element, only three (3) elements will undergo a secondary pressure drop sequence following cold start as specified in Figure 2.

4.6.5 Filtering media construction, immersion, and thermal cycling. Filter element filtering media construction shall not be affected by thermal cycling and pressure-flow cycling. Normal operation shall not cause continuous or sporadic release of the media, binder, and adhesive if used and shall not result in degradation of element below rated efficiency level requirements. Filter elements shall be immersed in hydraulic fluid for a period of 24 hours at a fluid temperature of 275 ºF and 24 hours at a fluid temperature of -65 ºF in two separate environmental chambers. This cycle shall then be repeated three more times for a total period of 192 hours (96 hours at a fluid temperature of 275 ºF and 96 hours at a fluid temperature of -65 ºF). Transfer between hot and cold chambers shall not exceed 10 minutes. It is very important to switch from one test chamber to the other as quickly as possible, but not to exceed 10 minutes. All internal parts shall be in contact with the test fluid during this period. After immersion, the filter shall remain in this fluid at room temperature until further tests are conducted.

4.6.6 Cold start. Twelve (12) of the thirteen (13) filter elements as listed in Figure 2, shall be installed in the test housing filled with hydraulic fluid and subjected to a temperature of -65º ± 5 ºF for four hours in a test setup like Figure 6. The elements shall then be subjected to 10 flow cycles at a differential pressure as specified in the applicable specification sheet across the element with fluid held at -65º ± 5 ºF or lower. Each flow cycle shall be 15 ± 1 seconds in duration. See Figure 6 for cold start cycle wave shape. A flow cycle-pressure time trace shall be prepared and included in the qualification test report. There shall be no evidence of filter media damage for each element as evidenced by visual examination and satisfactory completion of subsequent tests listed in Figure 2, or the applicable specification sheet. The test fluid use for this test shall be MIL-PRF-5606 or MIL-PRF-87257, which have similar viscosity at -65ºF.

4.6.7 Filter dynamic efficiency. Three (3) filter elements shall undergo the dynamic efficiency test as specified in ISO 16889 with modified test flow cycling parameter in a test set up as depicted in Figure 7. Filter elements shall meet the minimum performance requirements as specified in their applicable slash sheet. Mechanical vibration shall be performed during the dynamic efficiency test at a frequency of 120 ± 10 cps and double amplitude of 0.014 inch in a direction parallel to the bowl axis of the assembly or as specified within the applicable slash sheet. Documentation shall cover both high to low and then low to high particle count cyclic flow readings and report average filtration ratios as depicted in Figure 10, 11, 12 and 13. If the filter element has a protective device (relief valve), the protective device shall be disabled per 4.5.5.2.

4.6.7.1 Dynamic efficiency sequence. This test sequence is intended to be run to completion on the three elements (3 as determined in 4.6.3.4 based on lowest B/P ranking), prior to completing any other sequence, as the minimum B/P value is identified and will be required to complete other test events.

4.6.7.2 Dynamic efficiency test conditions. The filter elements shall be subjected to the following test conditions:



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Fluid Temperature: 160°F ± 15°F
Test Fluid: MIL-PRF-83282 only
Test Housing: Non-bypass Style B Test Housing
Flow Change Rate: 150 – 250 msec (See Figure 8)
Terminal Pressure: 300 psid or as specified in the applicable Specification Sheet
Test Dust: ISO Medium Test Dust (ISO 12103-1 A3)
Dilution: None
Upstream Gravimetric: 3 mg/l (Can be adjusted lower (1 mg/l) for smaller filters)

4.6.7.3 Cyclic frequency. The cyclic flow frequency shall consist of one cycle per 240 to 262 seconds, minimum, as depicted in Figure 8, and it shall be documented in qualification test report.

4.6.7.4 Particle counting (4, 5, 6, 7, 14 and 20 microns (c)). Using an automated particle counter, particle counting shall be synchronized with flow changes as depicted in Figure 8. At least one of the data points for upstream and downstream counts shall be taken after at least 7 seconds of changes in flow every flow cycle. Particle count readings shall not be taken during flow changes.

4.6.7.5 Dynamic efficiency test results. Dynamic efficiency test results shall be recorded in accordance with Figures 10 through 13. In addition, data shall be plotted showing filtration ratio ($\alpha_{X(c)}$) vs. particle size ($\mu_{(c)}$), differential pressure (psid) vs. time (s), filtration ratio ($\alpha_{X(c)}$) vs. differential pressure (psid), differential pressure (psid) vs. contamination (g), and filtration ratio ($\alpha_{X(c)}$) vs. percent test time. Test samples shall be required to meet efficiency ratios as defined in the applicable slash sheet.

4.6.8 Filter elements unloading. The filter elements shall be installed in a test filter housing as depicted in Figure 7 and loaded until 300 psid ± 5% is achieved across the element, or as specified within the applicable slash sheet. Pressure, flow, effluent particulate passing, and time shall be recorded in accordance with the applicable slash sheet. This loaded condition is considered the Terminal Differential Pressure and should be reached at the completion of the Dynamic Efficiency Test. Upon reaching the terminal differential pressure, the fluid shall be reduced to a pressure of 0 psi and a flow of 0 gpm. Record this pressure, flow, effluent particulate passing and time (minutes and seconds). The test filter shall remain at rest with no flow or pressure for a minimum of 20 seconds ± 10 seconds. Following this rest period, pressure and flow rate shall be increased back to the rated flow within 5 seconds. Pressure, flow, and time shall be maintained for 15-minutes, and the lowest (first peak) pressure drop shall be recorded. Measured pressure differential across the element and effluent particulate count passing/releasing from the element and document results. The average particle count shall be documented for each individual filter element and for the three test units, collectively. The filters tested shall not exceed 10% pressure drop from the first test point. This is a pass / fail criterion. Pressure drop (psid) vs. time (s) and particle count data for >4, >6, and >14 microns (c) shall be provided for each of the three elements tested and documented in the qualification test report.

4.6.9 Media migration. Filter elements shall withstand vibration frequencies IAW paragraph 4.6.9.2 below. There shall be no media migration traceable to filter element, media, its adhesives,



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or its packaging. Prior to start of media migration tests, and upon obtaining the fluid cleanliness requirements stated in 4.6.2.2, the test fluid from the test filter housing shall be filtered using a membrane filter 47 mm diameter disk, absolute 0.8-micron type AA Millipore Patch or equivalent. This patch will be the referee sample to be compared to the results of the filter element, media, and packaging migration test below. Note: This hydraulic fluid must be captured to use in procedures below.

4.6.9.1 Filter element, media, and packaging migration test. A sinusoidal vibration test shall be conducted on the filter assembly at room temperature in each of the filter assembly's three mutually perpendicular axes. One of these axes shall be that which is parallel to the exit flow path of the filter element. The filter assembly shall be mounted on a resonance-free fixture for test (see Figure 16). The vibration input accelerometer shall be mounted on the fixture close to the mounting point of the filter assembly to the fixture. A second accelerometer shall be mounted directly on the filter assembly for monitoring resonant frequencies. The rate of change of frequency shall be logarithmic.

Vibration tolerances
Amplitude ± 10%
Frequency ± 2%
Acceleration ± 10%

4.6.9.2 Vibration. The following vibration tests shall be conducted:

(a) Resonance search - resonant frequencies of the filter assembly shall be determined by varying the frequency of applied vibration slowly through the range specified in Table I, Part 1, at reduced levels but with sufficient amplitude to excite them. This shall be repeated for each of the three axes.

(b) Resonance dwell - the filter assembly shall be vibrated along each axis at the four most severe resonant frequencies (determined in the resonance search) for 30 minutes at each resonance. The vibration levels and test times shall be per Table I, Part 1.

(c) Cycling test - the filter assembly shall be vibrated along each axis in accordance with the test levels, frequency range and times from Table I, Parts 1 and 2. The frequency of applied vibration shall be swept over the frequency range of 5 to 2000 to 5 Hz logarithmically. The time to sweep from 5 to 2000 to 5 Hz shall be approximately 20 minutes. If a change in the resonant frequency occurs during the test, its time of occurrence shall be recorded and immediately the frequency shall be adjusted to maintain the peak resonance condition. The final resonant frequency shall be recorded and documented in the qualification test report.



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TABLE I. Vibration – Frequency - Time Schedule

| | Vibration level | Frequency Band, Hz | | | | |
|---|---|---|---|---|---|---|
| Part 1 | 0.1 inch double amplitude | 5 to 14 | | | | |
| | 1.0 g | 14 to 23 | | | | |
| | 0.036 inch double amplitude | 23 to 74 | | | | |
| | 10.0 g | 74 to 2000 | | | | |
| Part 2 | 0.2 inch double amplitude | 5 to 10 | | | | |
| | 1.0 g | 10 to 18 | | | | |
| | 0.06 inch double amplitude | 18 to 81 | | | | |
| | 20.0 g | 81 to 2000 | | | | |
| Vibration time schedule - (Time per one axis) | | | | | | |
| Part 1 | Number of resonances | 0 | 1 | 2 | 3 | 4 |
| | Total vibration time at each resonance | 0 | 0.5 hr | 1 hr | 1.5 hr | 2 hr |
| | Cycling Time | 2 hr | 1.5 hr | 1 hr | 0.5 hr | 0 |
| Part 2 | 30 minutes cycling per axis - no resonance dwell | | | | | |

4.6.9.3 Filter migration analysis. The lockwire, caps, and filter bowl shall be removed from the test housing and the test fluid, including fluid trapped inside the test filter, shall be filtered again using a membrane filter 47-mm diameter disk, absolute 0.8-micron type AA Millipore Patch or equivalent. This patch and what was captured after vibration, will be visually compared to the referee sample from paragraph 4.6.9 above, to ensure reliable results are achieved.

4.6.9.4 Filter element media migration results. After completion, filter patch membrane shall be examined for evidence of filter media migration using the procedure specified in SAE-AS598. A visual comparison shall be made of the referee sample (paragraph 4.6.9.3 above), coming from the test housing compared to the media migration of the filter element. There shall be no identifiable particulate as coming from the filter element, media, or packaging. A scanning electron microscope (SEM) with a minimum of 100x magnification shall be used to verify that no particulates can be traceable to the filter media. A minimum of 4 photograph(s) from the SEM shall be taken and documented in the qualification test report.

4.6.10 Flow fatigue. Filter elements shall be subjected to pressure-flow cycles with hydraulic fluid at minimum of 30% of rated flow and at 225ºF ± 5ºF. The test fluid shall be contaminated at a minimum of 5 mg/liter as a starting point and continue loading at lower concentrations until filter is totally loaded or maximum dP is reached. A cycle shall consist of increasing the differential pressure across the filter element from zero to the maximum specified pressure and back to zero by first increasing, then decreasing, the flow through the test filter



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element, which has been loaded with ISO medium test dust. Peak pressure shall be reached within 25% of the cycle and held for at least 50% of each cycle. The cycling rate and the number of cycles at each differential pressure shall be in accordance with Figure 9, (with Max psid as specified in the applicable specification sheet). There shall be no evidence of damage as indicated by the post B/P test. There shall be no by-passing of fluid. For this test the use of MIL-PRF-83282 is required.

4.6.10.1 Flow fatigue test sequence. This sequence of testing (Figure 2) is intended to be run on one filter element (1- article) with the seventh ($7^{th}$) lowest B/P as determined in 4.6.3.4.

4.6.11 Pressure element build-up and collapse sequence. Filter elements shall withstand a differential pressure as shown in Figure 2, on two (2) elements with the highest B/P and as specified in this document and the applicable specification sheet.

4.6.11.1 Pressure build-up and collapse for all elements. Pressure build-up and collapse shall be conducted in accordance with ISO 2941, using ISO Medium Test Dust as a contaminant and MIL-PRF-83282 as the test fluid. Qualification test samples shall be required to meet differential pressures as defined in the applicable slash sheet.

4.7 Markings.

4.7.1 Identification of product. Each filter element shall be clearly and permanently identified on one end by steel stamping, electro-etching, or by applicable laser marking. The method of marking shall be in accordance with MIL-STD-130. The marking shall not be placed on a sealing surface and the markings shall not affect the sealing of the element. The following information shall be provided:

M32746 (insert slash number of specification sheet and size dash number)
Manufacturer's part no
Manufacturer's lot no
Manufacturer's name or trademark

4.8 Conformance inspection. Conformance inspection shall consist of the following:

4.8.1 Individual inspection. The individual inspection shall be conducted on every filter element manufactured in the order specified in Figure 3. The filter element Minimum B/P shall be verified on 100% of the filters manufactured. Any filter element containing a defect shall be rejected.

4.8.2 Sampling lot inspections. The sampling lot inspection may be conducted on filter elements in the order specified in Figure 3. Quantity and recurring schedule should be based upon vendor quality procedures and processes and ANSI/ASQ Z1.4. Any of the filter elements containing a defect shall be reported to the qualifying activity immediately.



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## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of the materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Preservation and packaging. Unless otherwise specified in the contract, the preservation and packaging below is satisfactorily used in the past for filter elements.

5.2.1 Maximum protection. Maximum protection is the degree of preservation or packing required for protection of material against the most severe conditions known or anticipated to be encountered during shipment, handling, and storage. Preservation and packing so designated will be designed to protect material against direct exposure to extremes of climate, terrain, operational and transportation environments without protection other than that provided by the pack. The conditions to be considered include, but are not limited to:

a. Multiple handling during transportation and in transit storage from point of origin to ultimate user.

b. Shock, vibration, and static loading during shipment.

c. Loading on ship deck, transfer at sea, helicopter delivery, and offshore or over-the-beach discharge, to ultimate user.

d. Environmental exposure during shipment or during in transit operations where port and warehouse facilities are limited or nonexistent.

e. Extended open storage in all climatic zones.

f. Static loads imposed by stacking.

5.2.2 Preservation (elements). Unless otherwise specified in the contract, each element shall air dry after B/P testing (4.6.3.5). Air drying shall be accomplished in an atmospherically controlled clean room or laminar flow bench. When dry, the element shall then be placed within a heat-sealed bag conforming to MIL-DTL-117, Type I, Class E, Style 2.

## 6. NOTES

(This section contains information of a general or explanatory nature which may be helpful but is not mandatory.)



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6.1 Intended use. The filter elements covered by this specification are intended for use in aircraft and missile hydraulic systems covered by SAE-AS5440 and SAE-AS8775 operating with hydraulic fluid conforming to MIL-PRF-5606, MIL-PRF-83282, MIL-PRF-87257 at nominal operating pressures 1,500 psi to 5,000 psi and temperature range -65°F to +275°F.

6.1.1 Military unique application. The filter elements covered by this specification are specifically designed and dimensioned uniquely for military aircraft and missile hydraulic systems as specified herein and in the specification sheets and not available for use with commercial aircraft or other non-military equipment systems.

6.2. Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. PIN - part identification number (See MIL-DTL-32746/XX).
c. Element type, class, micron rating, and size (see 1.2).
d. Qualified Product Database (QPD) and activity responsible (US Army) for qualification inspection.
e. Packaging and Qualification (see sections 5.1 and 6.3).

6.3 Qualification. With respect to products requiring qualification such as this filter element (see 4.2.2), awards will be made only for such products as have, prior to the time set for opening of bids, been tested, and approved for inclusion in the applicable Qualified Products List (QPL), whether such products have been so listed by that date. The attention of suppliers is called to this requirement, and others listed below, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the QPL is the US Army Futures Command, DEVCOM Aviation and Missile Center, Attn: Technical Data Management Division (TDMD), Building 5687, 5687 Logistics Road, Redstone Arsenal, AL 35898-5000

6.4 Subject term (keyword) listing.

Army Filters
Cold start
Hydraulics
Media Migration



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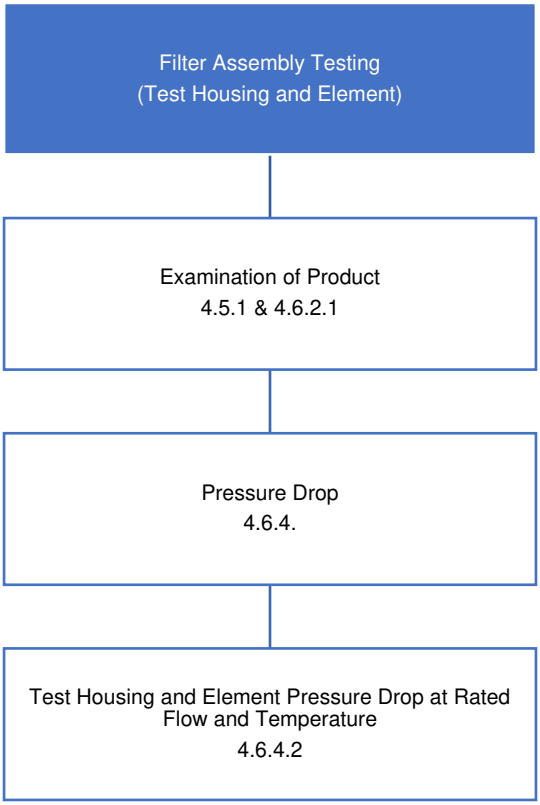



FIGURE 1. Qualification Inspections / Test



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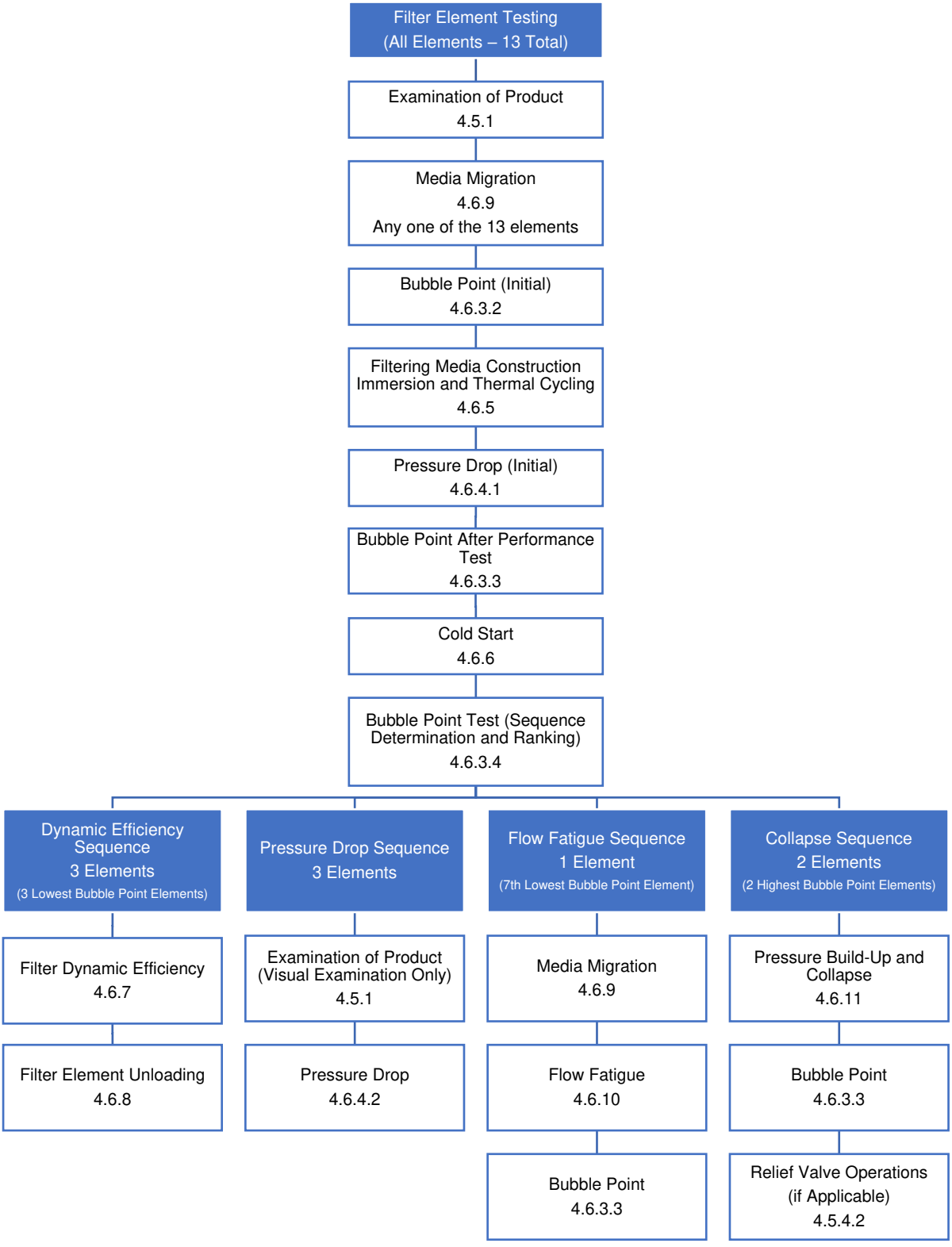



FIGURE 2. Qualification Inspections / Test (Filter Element Only).



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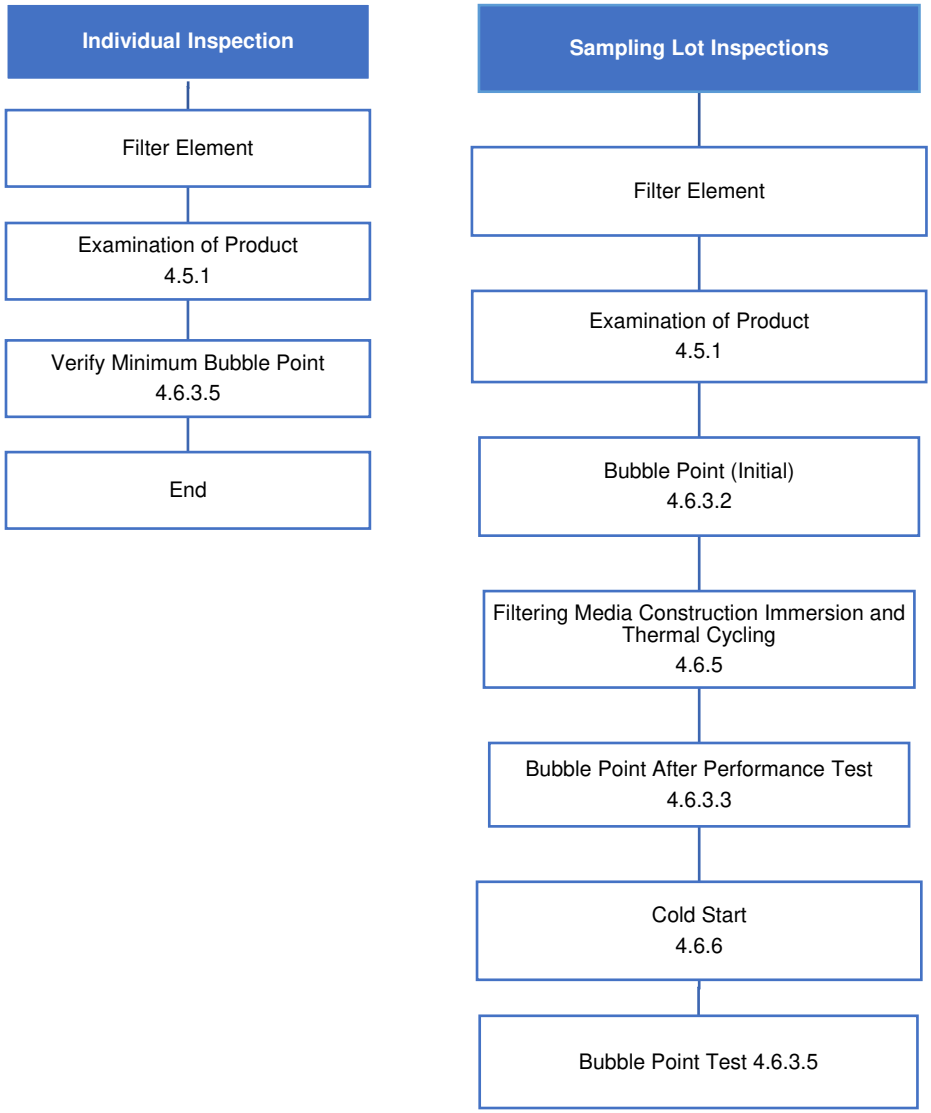



FIGURE 3. Conformance Inspections (Individual Inspection and Sampling Lot Inspections)



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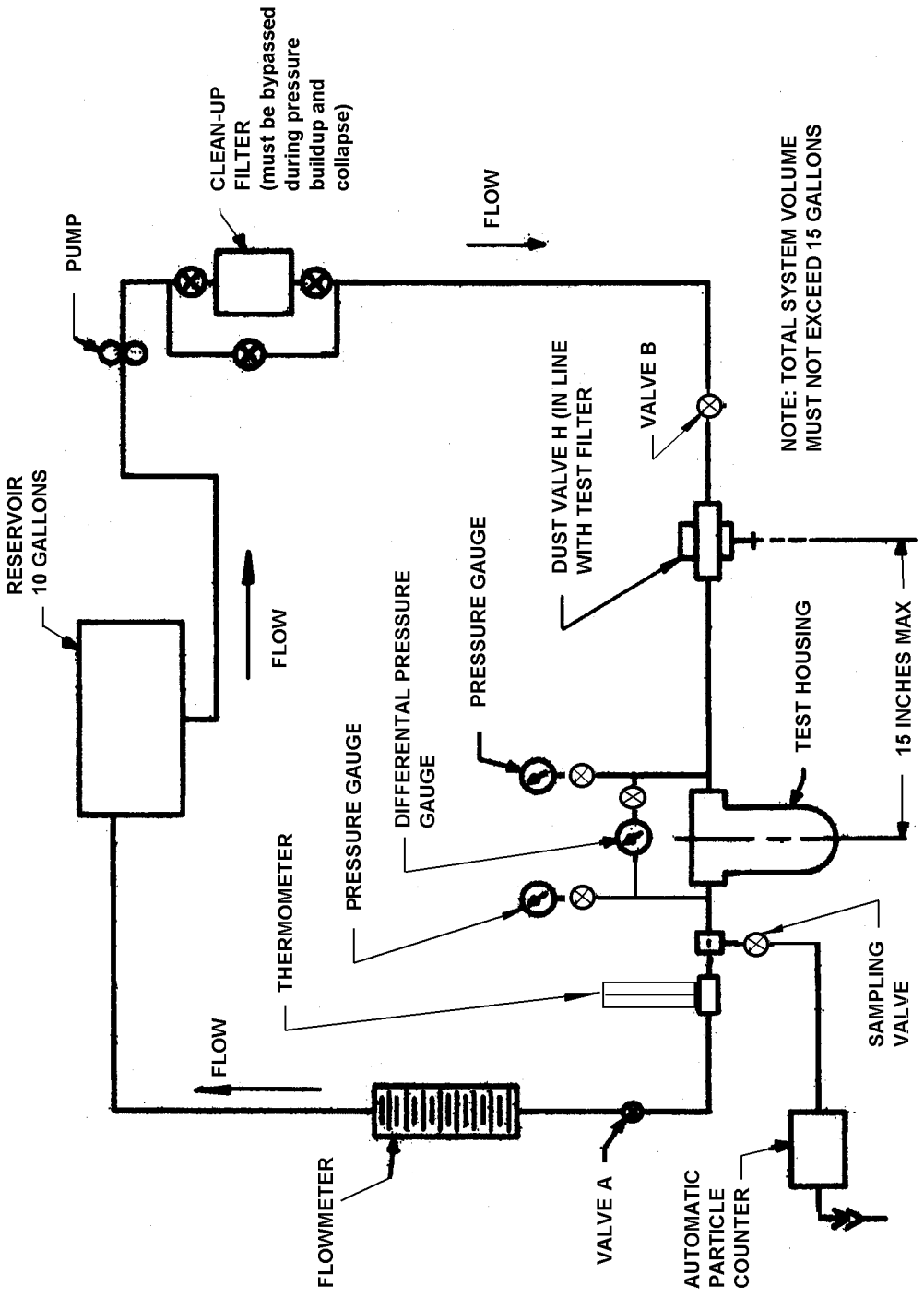



FIGURE 4. Typical setup for determining pressure buildup and collapse pressure characteristics



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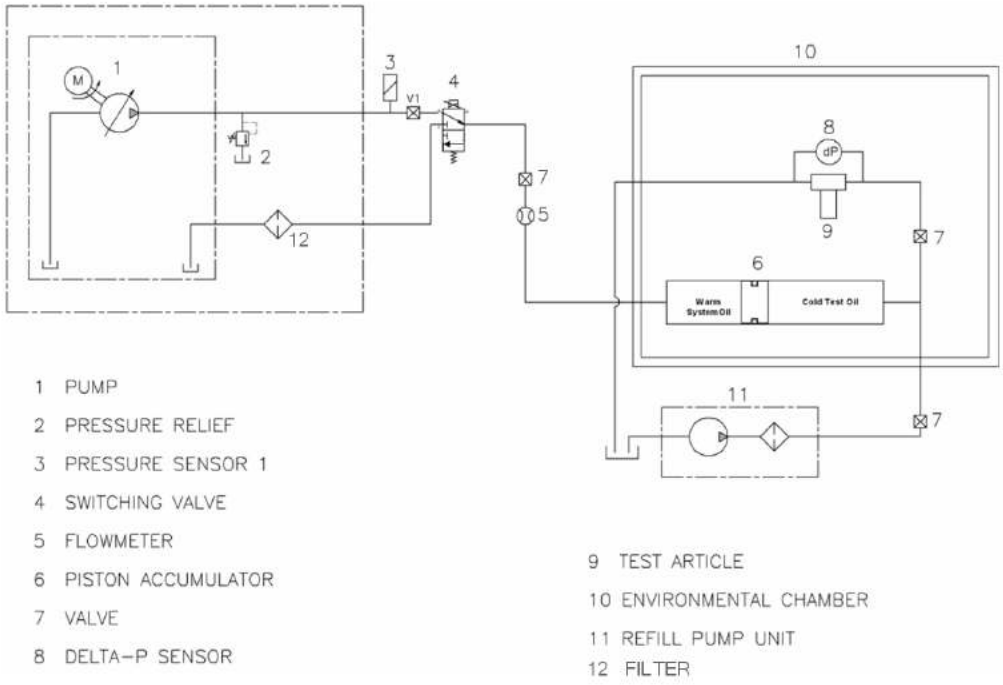



FIGURE 5. Typical schematic diagram for cold-start test

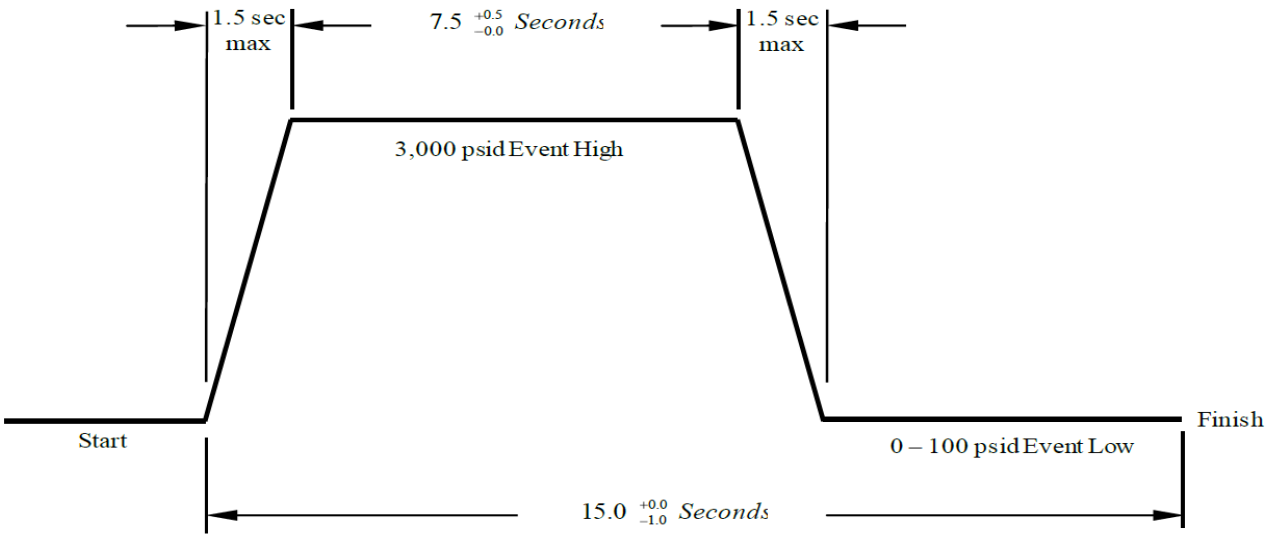
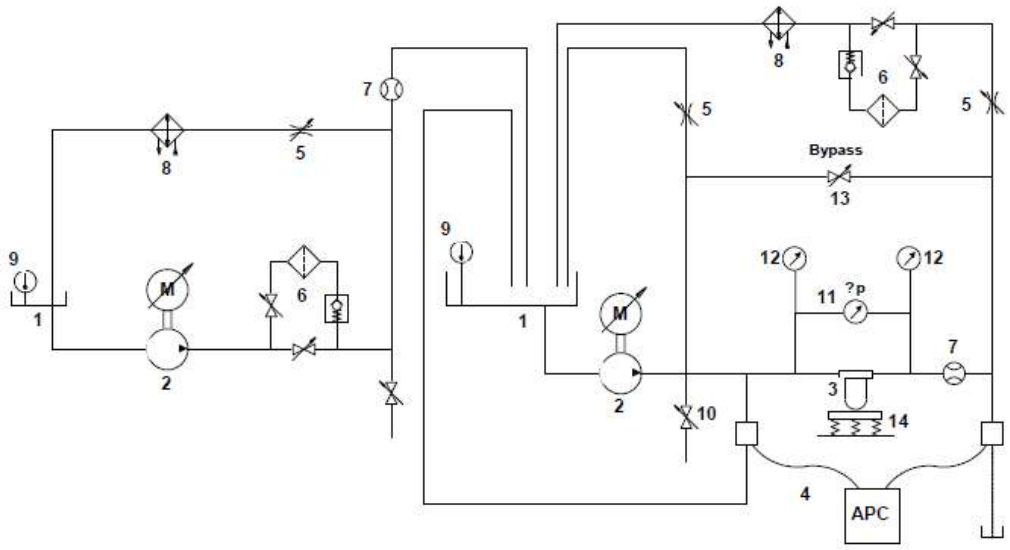



FIGURE 6. Cold start cycle wave shape



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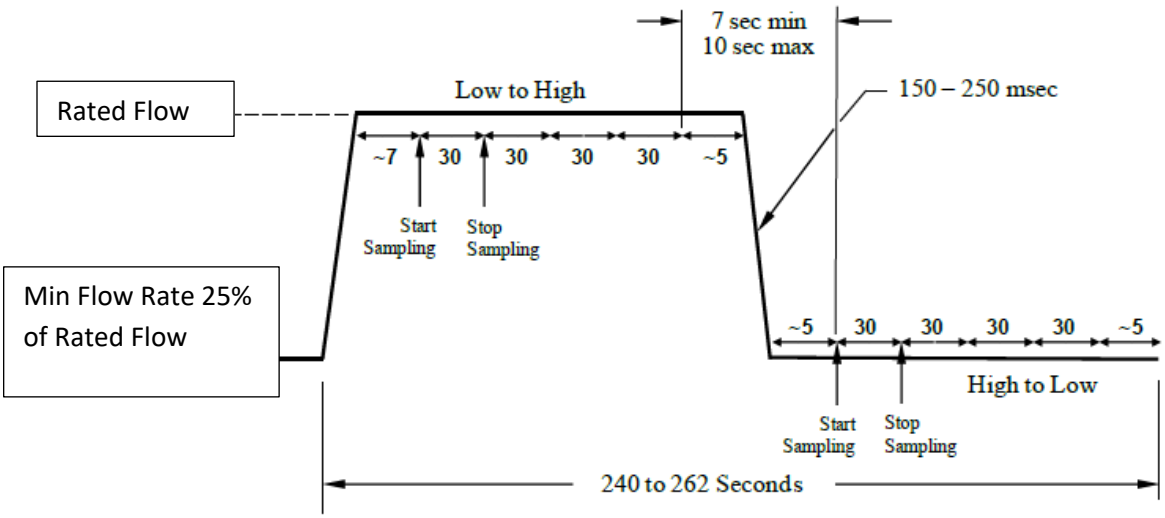

| | |
|---|---|
| 1 Reservoir | 8 Heat Exchanger |
| 2 Pump | 9 Temperature Sensor |
| 3 Unit Under Test | 10 Sampling Valve |
| 4 Particle Counting System | 11 Differential Pressure Indicator |
| 5 Regulating Valve | 12 Pressure Gauge |
| 6 Cleanup Filters | 13 Cyclic Flow Bypass Valve |
| 7 Flow Meter | 14 Vibration Table |

Note: The test set up shall use either a hydraulic pump with a variable speed motor (2) or a cyclic flow bypass valve (13). The option is at the discretion of the facility performing the test.

FIGURE 7. Typical schematic diagram for dynamic efficiency test



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**Note:** Delay time between count periods is a function of particle counter design – can be 0 – 5 seconds.

FIGURE 8. Particle count synchronization for dynamic efficiency test



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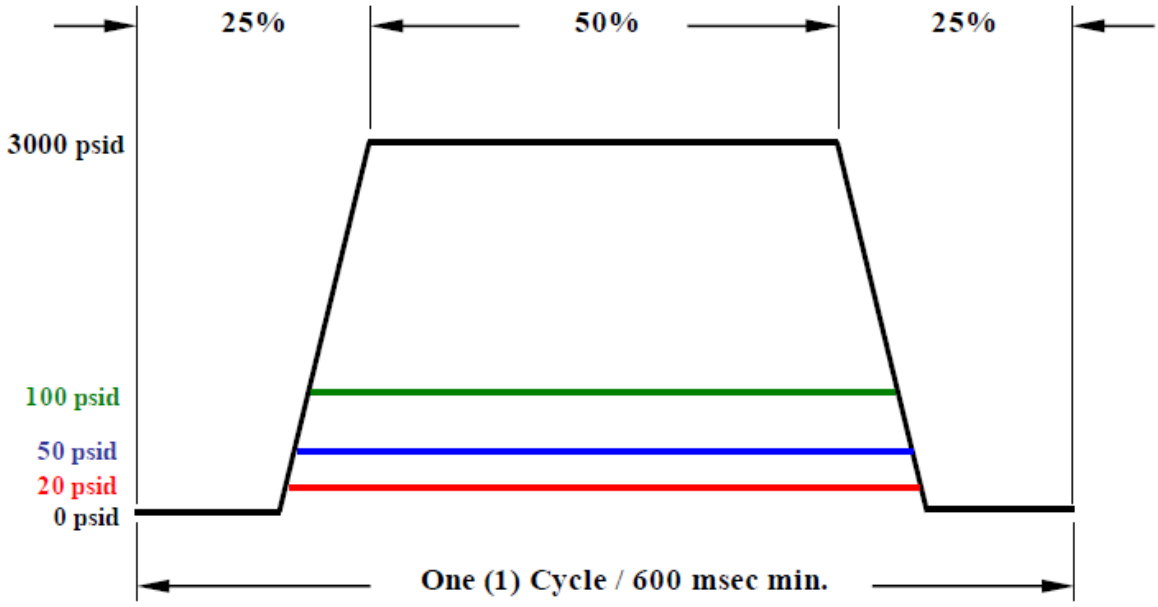



FIGURE 9. Cyclic frequency for flow fatigue test



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| Test Report | | | | | | |
|---|---|---|---|---|---|---|
| Test Number | Test Date | Operator | | | | |
| **FILTER AND ELEMENT IDENTIFICATION** | | | | | | |
| Element ID: | Housing ID: | | | | | |
| Spin on: | Minimum element bubble point: | | | | | |
| **OPERATION CONDITIONS** | | | | | | |
| **Test Fluid** | | | | | | |
| Type: | Ref: | Batch No: | | | | |
| Viscosity at the test temperature ($mm^2/s$): | Test temperature range (°C): | | | | | |
| Antistatic: | Type: | Conductivity (pS/m): | | | | |
| **Test Contaminant** | | | | | | |
| Type: | Batch No: | | | | | |
| **Test System** | | | | | | |
| Flow rate $^{q}$max (L/min): | Initial volume (L): | | | | | |
| Base upstream concentration $G_b$ (mg/L): | Final volume (L): | | | | | |
| **Injection System** | | | | | | |
| *Injection parameter* | *Initial* | *Final* | *Average injection parameters* | | | |
| System volume (L): | | | Injection flow $^{q}_{i}$ (L/min): | | | |
| Concentration (mg/L): | | | Concentration $G_i$ (mg/L): | | | |
| **Counting System** | *Counter and sensor* | *Flow Rate (mL/min):* | *Dilution ratio* | | | |
| Upstream | | | | | | |
| Downstream | | | | | | |
| Counter calibration | Method: | Date: | | | | |
| **TEST RESULTS** | | | | | | |
| **Element Integrity** | | | | | | |
| Bubble point to ISO 2942 | Wetting fluid: | | | | | |
| **Differential Pressure** | | | | | | |
| Initial ass'y, $\Delta p$ (kPa): | | | | | | |
| Final element, $\Delta p$ (kPa): | | | | | | |
| **Retention Capacity** | | | | | | |
| ISO MTD mass injected $M_i$ (g): | ISO MTD retained capacity $C_R$ (g): | | | | | |
| 80% upstream concentration $G_{80}$ (mg/L): | | | | | | |
| **Filtration Ratio $\alpha_{x(c)}$** | | | | | | |
| Average Filtration Ratio | | | | | | |
| Particle size, µm(c) | | | | | | |

FIGURE 10. Test report for dynamic efficiency test



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Differential Pressure Versus Contaminant Added

| Time Interval | Test Time (min) | Element Δp (kPa) | Injected mass (g) | Time Interval | Test Time (min) | Element Δp (kPa) | Injected mass (g) |
|---|---|---|---|---|---|---|---|
| 0% | | | | 60% | | | |
| 10% | | | | 70% | | | |
| 20% | | | | 80% | | | |
| 30% | | | | 90% | | | |
| 40% | | | | 100% | | | |
| 50% | | | | | | | |

Grand Average Filtration ($\alpha_{(c)}$) Ratio Report

| Time Interval | d>4 µm(c) | ratio α | d>5 µm(c) | ratio α | d>6 µm(c) | ratio α | d>7 µm(c) | ratio α | d>10 µm(c) | ratio α |
|---|---|---|---|---|---|---|---|---|---|---|
| Initial up | | | | | | | | | | |
| 10% Up | | | | | | | | | | |
| Down | | | | | | | | | | |
| 20% Up | | | | | | | | | | |
| Down | | | | | | | | | | |
| 30% Up | | | | | | | | | | |
| Down | | | | | | | | | | |
| 40% Up | | | | | | | | | | |
| Down | | | | | | | | | | |
| 50% Up | | | | | | | | | | |
| Down | | | | | | | | | | |
| 60% Up | | | | | | | | | | |
| Down | | | | | | | | | | |
| 70% Up | | | | | | | | | | |
| Down | | | | | | | | | | |
| 80% Up | | | | | | | | | | |
| Down | | | | | | | | | | |
| 90% Up | | | | | | | | | | |
| Down | | | | | | | | | | |
| 100% Up | | | | | | | | | | |
| Down | | | | | | | | | | |
| **Avg Up** | | | | | | | | | | |
| **Avg Down** | | | | | | | | | | |

FIGURE 11. Grand average filtration report for dynamic efficiency test



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:42Z
Check the source to verify that this is the current version before use.

**Dynamic Efficiency Test Report Average Filtration Ratio (AFR) low to high**

| |
|---|
| Operating Conditions |
| Record Time to cleanup: to ISO code level on preconditioned element – ISO 4406 Code 13/10/7 or below: |
| Cycle High Flow Interval: |
| Cycle Low Flow Interval: |
| Flow Rate: Low  Final Δp element: |
| High |
| Operating Temperature: |
| Vibration: |
| Flow Rise rate: |

**Retention Capacity**

ISO MTD mass injected $M_i$ (g):  ISO MTD retained capacity $C_R$ (g):

80% upstream concentration $G_{80}$ (mg/L):

**Filtration Ratio**

| | | | | | | |
|---|---|---|---|---|---|---|
| Average filtration ratio | | | | | | |
| Particle size, µm(c) | | | | | | |

**Particle counts per mL and Filtration Ratio – Average Filtration Ratio (AFR) low to high**

| Time Interval | d>4 µm(c) | ratio α | d>5 µm(c) | ratio α | d>6 µm(c) | ratio α | d>7 µm(c) | ratio α | d>10 µm(c) | ratio α |
|---|---|---|---|---|---|---|---|---|---|---|
| Initial Up | | | | | | | | | | |
| 1 event Up | | | | | | | | | | |
| Down | | | | | | | | | | |
| 2 event Up | | | | | | | | | | |
| Down | | | | | | | | | | |
| 3 event Up | | | | | | | | | | |
| Down | | | | | | | | | | |
| 4 event Up | | | | | | | | | | |
| Down | | | | | | | | | | |
| 5 event Up | | | | | | | | | | |
| Down | | | | | | | | | | |
| 6 event Up | | | | | | | | | | |
| Down | | | | | | | | | | |
| 7 event Up | | | | | | | | | | |
| Down | | | | | | | | | | |
| Avg. Up | | | | | | | | | | |
| Avg. Down | | | | | | | | | | |

FIGURE 12. Dynamic efficiency test report – Average Filtration Ratio (AFR) low to high



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:42Z
Check the source to verify that this is the current version before use.

**Dynamic Efficiency Test Report Average Filtration Ratio (AFR) high to low**

Operating Conditions

Record Time to cleanup: to ISO code level on preconditioned element – ISO 4406 Code 13/10/7 or below:

Cycle High Flow Interval:

Cycle Low Flow Interval:

Flow Rate: Low — Final Δp element:

High

Operating Temperature:

Vibration:

Flow Rise rate:

**Retention Capacity**

ISO MTD mass injected $M_i$ (g): ISO MTD retained capacity $C_R$ (g):

80% upstream concentration $G_{80}$ (mg/L):

**Filtration Ratio**

| | | | | | | |
|---|---|---|---|---|---|---|
| Average filtration ratio | | | | | | |
| Particle size, µm(c) | | | | | | |

**Particle counts per mL and Filtration Ratio – Average Filtration Ratio (AFR) high to low**

| Time Interval | d>4 µm(c) | ratio α | d>5 µm(c) | ratio α | d>6 µm(c) | ratio α | d>7 µm(c) | ratio α | d>10 µm(c) | ratio α |
|---|---|---|---|---|---|---|---|---|---|---|
| Initial Up | | | | | | | | | | |
| 1 event Up | | | | | | | | | | |
| Down | | | | | | | | | | |
| 2 event Up | | | | | | | | | | |
| Down | | | | | | | | | | |
| 3 event Up | | | | | | | | | | |
| Down | | | | | | | | | | |
| 4 event Up | | | | | | | | | | |
| Down | | | | | | | | | | |
| 5 event Up | | | | | | | | | | |
| Down | | | | | | | | | | |
| 6 event Up | | | | | | | | | | |
| Down | | | | | | | | | | |
| Avg. Up | | | | | | | | | | |
| Avg. Down | | | | | | | | | | |

FIGURE 13. Dynamic efficiency test report – Average Filtration Ratio (AFR) high to low



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:42Z
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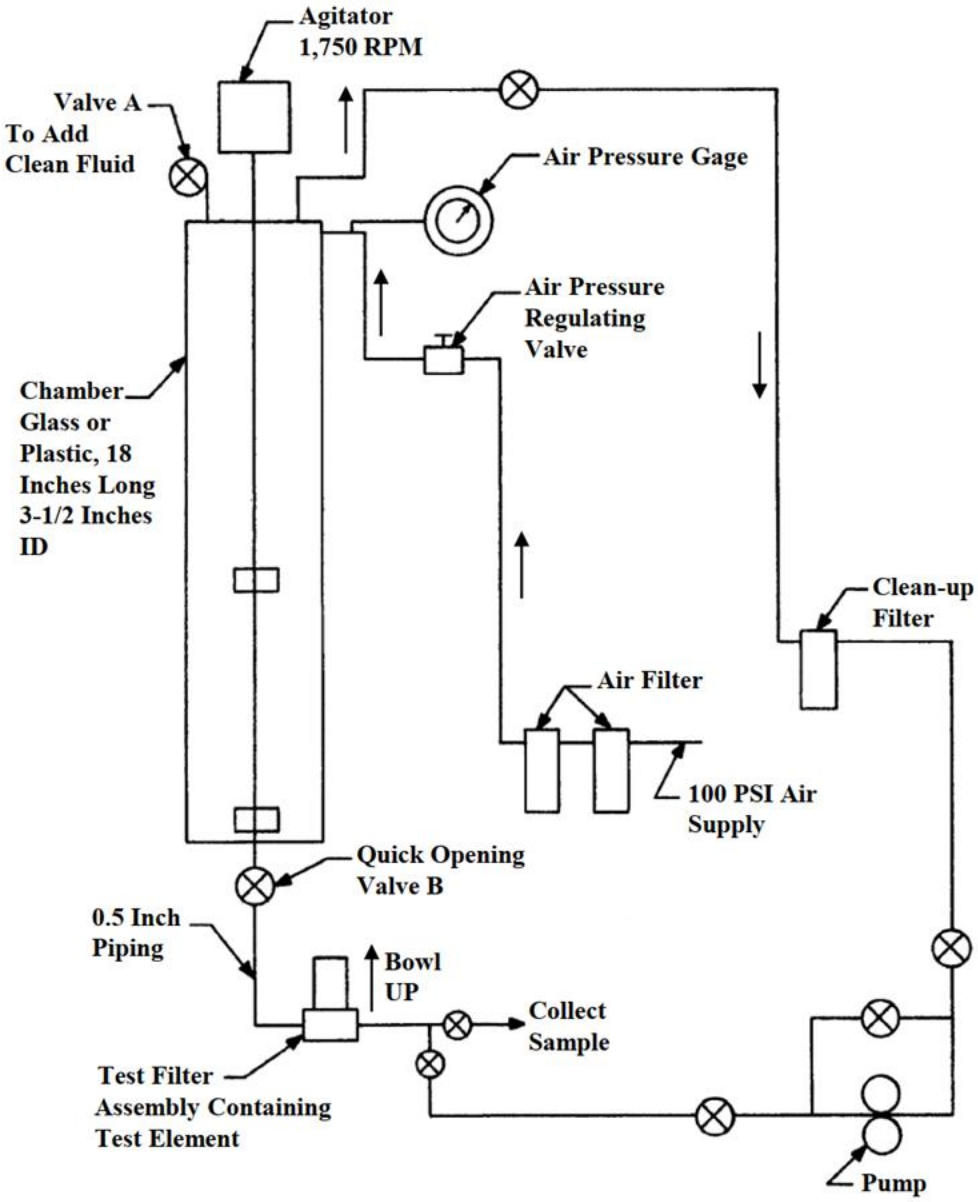



FIGURE 14. Media Migration Initial Set-up



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:42Z
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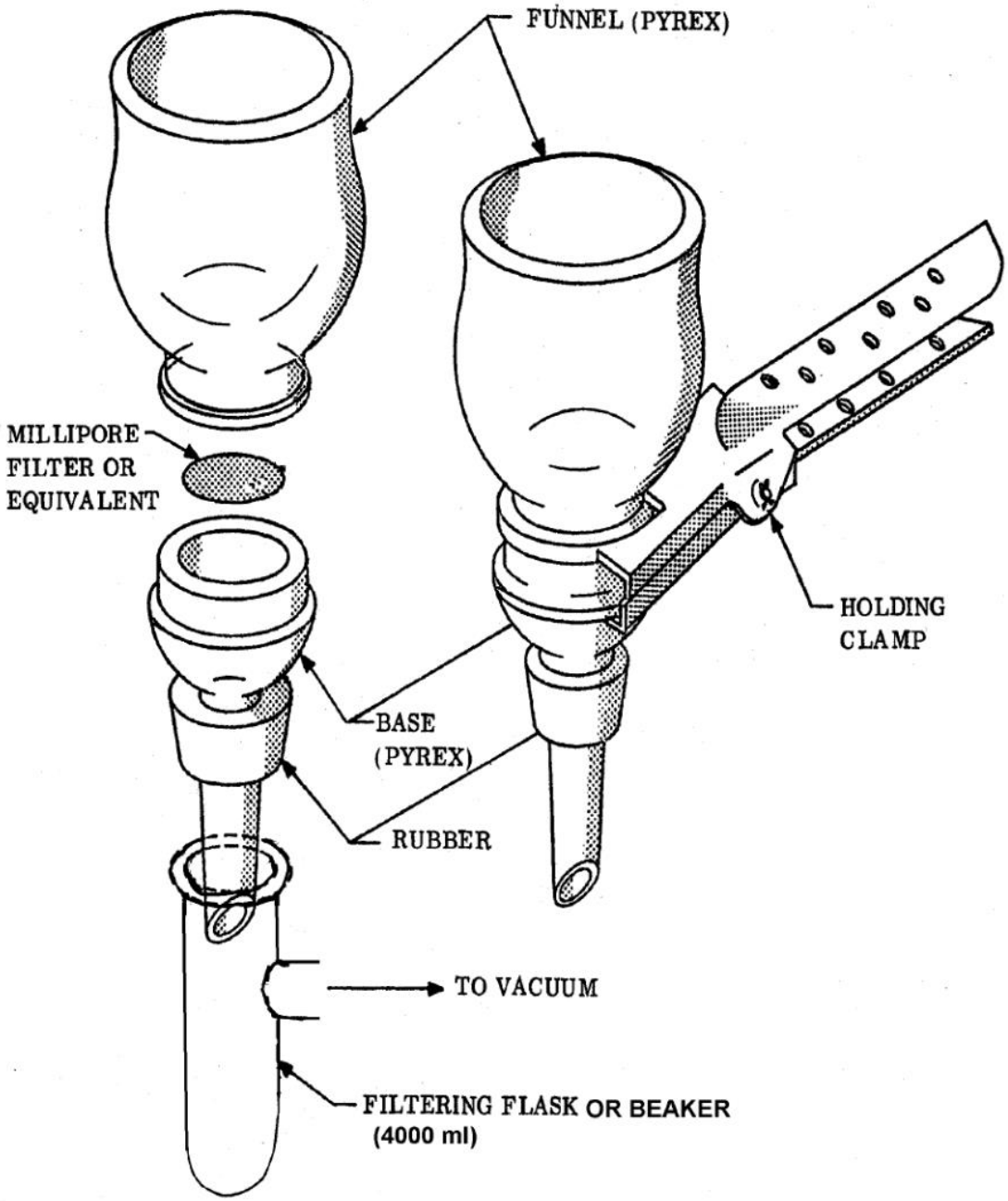
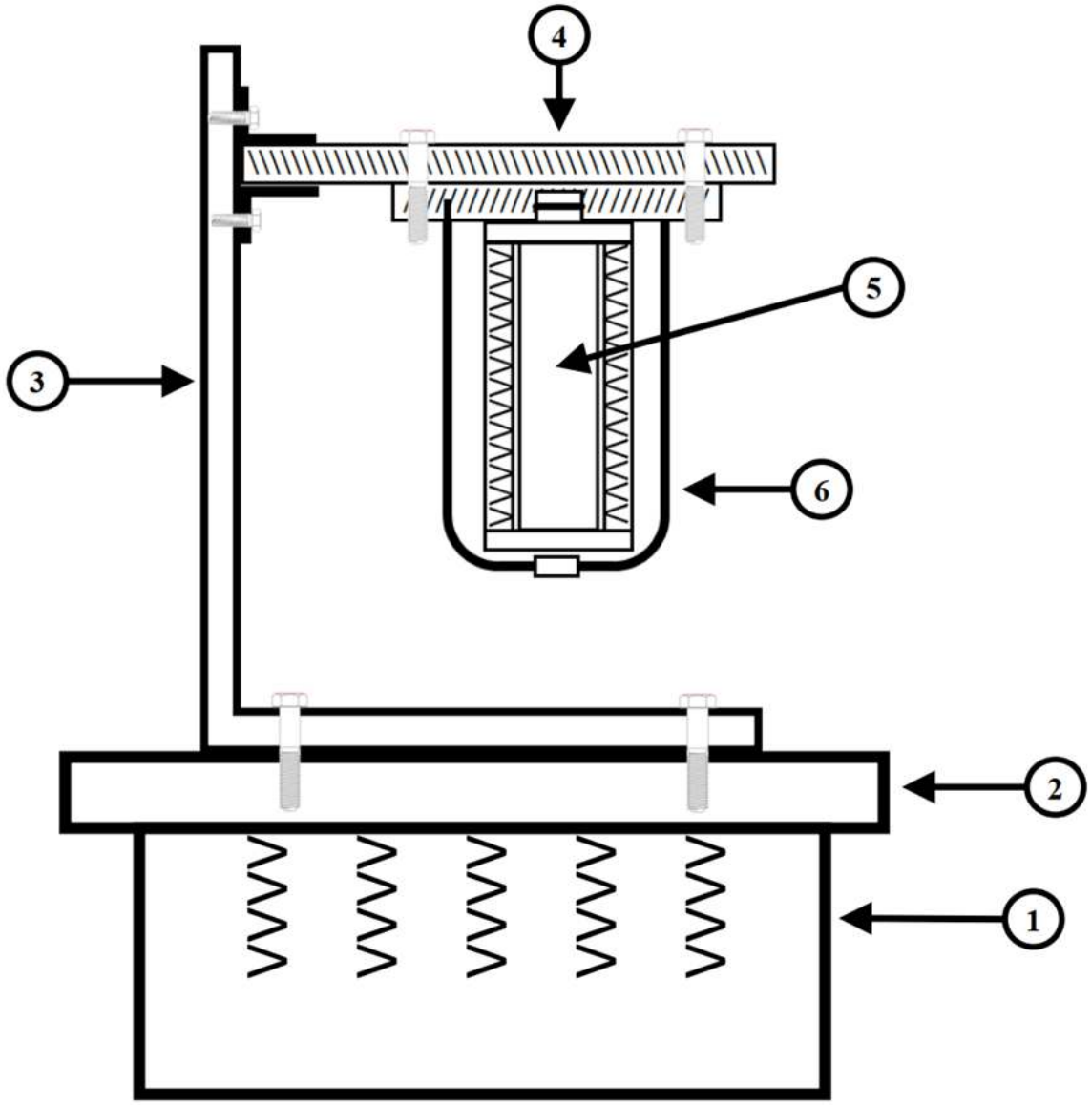



FIGURE 15. Media Migration Filter Patch Set-up



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1. Vibration Shaker Base
2. Vibration Table
3. Vibration Sturdy Mounting Fixture (Bracket)
4. Filter Mounting Bracket
5. Filter Element Under Test
6. Filter Bowl (Top & Inlets Capped Off)

FIGURE 16. Media Migration Set-up with Vibration



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CONCLUDING MATERIAL

Custodians:
Army - AV
Air Force -11

Preparing activity:
Army - AV
(Project 1650-2023-003)

Review activities:
Army – MI

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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